

tf.compat.v1.math.reduce_mean Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math/reduce_mean

Computes the mean of elements across dimensions of a tensor.

Function Signatures

```
tf.compat.v1.math.reduce_mean( input_tensor, axis=None,
keepdims=None, name=None, reduction_indices=None,
keep_dims=None )
```

Code Examples

```
tf.compat.v1.math.reduce_mean(
input_tensor,
axis=None,
keepdims=None,
name=None,
reduction_indices=None,
keep_dims=None
)
```

```
tf.compat.v1.math.reduce_mean(  
input_tensor,  
axis=None,  
keepdims=None,  
name=None,  
reduction_indices=None,  
keep_dims=None  
)
```

```
x = tf.constant([[1., 1.], [2., 2.]])  
tf.reduce_mean(x)  
  
tf.reduce_mean(x, 0)  
  
tf.reduce_mean(x, 1)
```

```
x = tf.constant([[1., 1.], [2., 2.]])
```

```
tf.reduce_mean(x)
```

```
tf.reduce_mean(x, 0)
```

Documentation

Computes the mean of elements across dimensions of a tensor.

Reduces `input_tensor` along the dimensions given in `axis` by computing the mean of elements across the dimensions in `axis`.

Unless `keepdims` is true, the rank of the tensor is reduced by 1 for each the entries in `axis`, which must be unique. If `keepdims` is true, the reduced dimensions are retained with length 1.

If `axis` is `None`, all dimensions are reduced, and a tensor with a single element is returned.

For example:

Returns

numpy compatibility

Equivalent to `np.mean`

Please note that `np.mean` has a `dtype` parameter that could be used to specify the output type. By default this is `dtype=float64`. On the other hand, `tf.reduce_mean` has an aggressive type inference from `input_tensor`, for example: